

BUS63-9

# Investment Philosophies

Value, contrarian, activist, growth, momentum, arbitrage, market timing, and passive — each tested against the evidence, costs, and biases that decide whether it survives

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Given what we actually know about how markets work, what kind of investor should you be — and can you defend it?

Albert School — 22 hours

## Preface

By now you can price a company. From **Corporate Finance** (BUS33-3) you discount cash flows, build a cost of capital from CAPM and a debt yield, and reason about value creation through the gap between **ROIC** and **WACC**. From **Valuation & Investment Analysis** (BUS43-3) you build a DCF, triangulate it against multiples, stress the two assumptions that move the number, and write a buy / hold / sell thesis you can defend. You can answer the question **what is this worth?**

This course asks the question that sits one level above the valuation: **given what we know empirically about how markets behave, what kind of investor should you be?** The two are not the same. A flawless DCF is useless if the market never closes the gap you identified, or closes it so slowly that costs and taxes eat the return, or if the “cheapness” you found is a value trap the rest of the market has correctly priced. An investment philosophy is the layer of belief that sits between **I can value things** and **I can make money** — a coherent set of convictions about where markets are inefficient, why those inefficiencies persist, and why **you** in particular are positioned to harvest them.

This is the conceptual complement to **Financial Instruments** (BUS63-8), which builds the quantitative toolkit of the Financial Markets track. There you learn the machinery; here you learn what to believe about the machinery and why. The distinction matters because every philosophy in this book — value, contrarian, activist, growth, momentum, arbitrage, market timing, passive — **works on paper**. Each has academic studies showing abnormal returns. The discipline this course builds is the habit of asking the three questions that separate a paper strategy from a live one: **Does the evidence survive once you subtract transaction costs and taxes? Does it survive the behavioural biases that make it psychologically hard to execute? And if it is real, why has it not been arbitrated away — what is your durable edge?**

The structure throughout is deliberate. We begin with **market efficiency**, because every philosophy is implicitly a bet that some form of efficiency fails. Then we test each philosophy in turn against the same three filters — **evidence, cost, bias** — never letting a strategy off the hook because its back-test looked good. We close where the course’s assessment closes: with you constructing and defending a **personal investment philosophy** coherent with your own risk tolerance, time horizon, tax status, and honestly identified informational edge. The goal is not to make you a value investor or a passive investor. It is to make you an investor who can say **what I believe about markets, why I believe it, and why it will still be true after costs** — and defend that worldview against someone paid to attack it.

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# 1 What an investment philosophy is — and why you need one

## 1.1 Two investors, one stock, opposite trades

A regional bank's shares fall 40% in a week after a credit scare. Two professional investors look at the same stock the same afternoon. The first **buys**: the franchise is intact, the market has overreacted to a headline, and at this price she is paid handsomely to wait for the panic to fade. The second **sells** (or shorts): a falling price is the market's verdict, the downtrend has momentum, and she will not step in front of it until the chart stabilises. Both are experienced, both are rational, and they have taken **opposite** sides of the same trade.

They disagree not about the facts but about **how markets work**. The first believes prices overreact to bad news and mean-revert; the second believes recent price trends persist. Each has an **investment philosophy** — a coherent worldview about market behaviour — and each trade follows logically from it. Strip the philosophy away and you have no basis for either trade: just two people guessing.

## 1.2 Definition and why coherence is the point

**Definition 1 (investment philosophy)** An **investment philosophy** is a coherent set of beliefs about **how markets work, where and why they misprice assets, and how those mispricings can be exploited**, from which a consistent investment **strategy** follows. The philosophy is the worldview; the strategy is its implementation; the **portfolio** is the strategy made concrete.

The word that does the work is **coherent**. An investor without an articulated philosophy does not hold no beliefs — they hold **inconsistent** ones, switching from value to momentum to market-timing as each fails, always arriving late and leaving late. The practical value of writing your philosophy down is that it tells you, in advance, **which opportunities are not yours to take and when a losing position is consistent with your thesis versus a signal that your thesis was wrong**.

### **Definition 2 (philosophy vs strategy vs tactics)**

- **Philosophy** — the belief about market behaviour (e.g. “markets overreact to bad news”). Stable for years.
- **Strategy** — the systematic way to exploit it (e.g. “buy the 10% worst-performing large caps each year, hold three years”). Stable across cycles.
- **Tactics** — the individual decisions (e.g. “buy this bank today at this price”). Change constantly.

**Common pitfall** The most expensive mistake an investor makes is not picking the **wrong** philosophy — it is having **none**, and therefore switching philosophies at the worst possible moment: abandoning value after a bad value year (just before it works), chasing momentum after the momentum has run, capitulating to passive at the market bottom. A mediocre philosophy **consistently applied** beats a sequence of excellent philosophies each abandoned at its low. Coherence over time is itself an edge.

## 1.3 Matching the philosophy to the investor: risk, horizon, tax

A philosophy is not chosen in the abstract; it must fit the **investor**. The same strategy that is right for a 30-year-old with stable income and a 30-year horizon is wrong for a foundation that must spend 5% of assets every year, and wrong again for a taxable investor in a high bracket.

**Definition 3 (the three matching dimensions)** A philosophy must be coherent with the investor's:

- **Risk tolerance** — both the financial **capacity** to bear losses and the psychological **willingness** to hold through them. A strategy you abandon in a drawdown has a real return of zero, however good its back-test.
- **Time horizon** — how long capital can stay committed. Strategies that exploit slow mean-reversion (value, contrarian) need years; an investor who may need the cash in 18 months cannot run them.
- **Tax status** — taxable versus tax-exempt, and the bracket. High-turnover strategies that look strong pre-tax can be destroyed by short-term capital-gains tax (Chapter 12).

**Worked example — Same edge, wrong investor.** A contrarian deep-value strategy historically earns strong returns but does so through **long, painful** periods of underperformance before the reversal — and high portfolio turnover when positions finally re-rate. It fits a tax-exempt endowment with a 20-year horizon and a board that will not panic. It is **actively dangerous** for a 64-year-old taxable investor two years from drawdown: the horizon may end inside a drawdown, and the turnover generates taxable gains. Same evidence, opposite verdict — because the investor is different.

**Common pitfall** Risk tolerance is discovered in drawdowns, not questionnaires. Investors routinely overstate their tolerance in calm markets and discover the truth at the bottom, where they sell. Design your philosophy around the risk you will **actually** hold through, not the risk you imagine holding through on a good day.

## 2 Market efficiency: the question every philosophy answers

### 2.1 Why this chapter comes first

Every active philosophy in this book is a bet that the market gets **some** prices wrong in **some** predictable way. So the first question is not “which strategy?” but “how efficient is the market, and where does its efficiency fail?” If markets were perfectly efficient, the only rational philosophy would be passive indexing (Chapter 11). Every active philosophy is therefore, at root, a specific claim about **which form of efficiency breaks**.

### 2.2 A concrete test before the theory

A new earnings report comes out at 8:00 a.m.: profits far above expectations. If you could buy at 8:01 and reliably earn an abnormal return as the price drifts up over the next week, the market would be failing to incorporate **public** information instantly. Whether that drift actually exists is an empirical question (it does — it is **post-earnings-announcement drift**, Chapter 8). Market-efficiency theory is just the disciplined vocabulary for asking, of any such pattern, **which information is already in the price?**

## 2.3 The three forms of the Efficient Market Hypothesis

**Definition 4 (Efficient Market Hypothesis (EMH))** A market is **efficient** with respect to an information set if prices fully and instantly reflect that information, so no strategy using only that information can earn **risk-adjusted abnormal returns**. Three nested forms differ by the information set:

- **Weak form** — prices reflect all **past price and volume** data. If true, **technical analysis** and momentum cannot work.
- **Semi-strong form** — prices reflect all **publicly available** information (prices plus fundamentals, news, filings). If true, **fundamental analysis** on public data cannot earn abnormal returns — this kills most value and growth strategies.
- **Strong form** — prices reflect **all** information, public **and** private. If true, even **insider** information cannot earn abnormal returns.

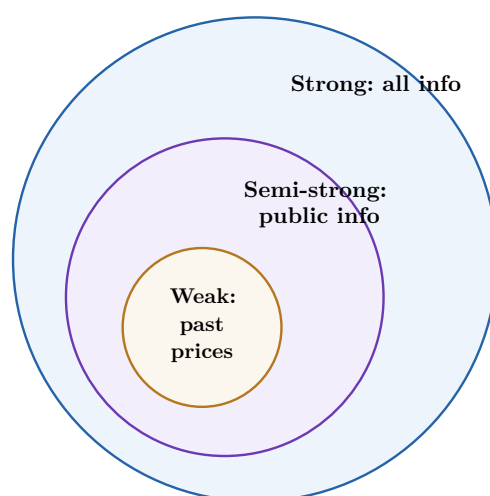


Figure 1: The three EMH forms are **nested**: strong implies semi-strong implies weak. Each active philosophy is a claim that one ring fails — momentum bets the weak form fails, value and growth bet the semi-strong form fails, insider trading bets the strong form fails. The empirical consensus: the weak and semi-strong forms hold **approximately** but not perfectly, and the strong form clearly fails.

**Remark** Efficiency is always **relative to a risk model**. An “abnormal return” is a return above what the risk taken should earn — so every efficiency test is a **joint test** of efficiency **and** the model used to define normal returns (this is the “joint hypothesis problem”). A pattern that looks like inefficiency may just mean your risk model is wrong. Keep this caveat live: it is the single most important reason to distrust a clean-looking anomaly.

## 2.4 Testing a strategy against the evidence: event studies and portfolios

**Definition 5 (two ways to test a strategy)**

- **Event study** — measure the **abnormal return** (actual minus expected) in a window around a specific event (earnings, split, acquisition). Answers: **how fast and how completely** does the price react to this information?

- **Portfolio approach** — sort the universe on a characteristic (e.g. P/E), form portfolios, and compare their **risk-adjusted** returns over long horizons. Answers: **does this characteristic predict returns?**

## 2.5 Why back-tests lie: the three killers

A strategy that earned 18% a year in a back-test routinely earns far less — or nothing — when traded live. Three forces explain almost all of the gap, and recognising them is the core skill of **learning outcome 2**.

**Definition 6 (survivorship bias)** **Survivorship bias** is the distortion from testing a strategy only on the firms (or funds) that **survived** to the end of the sample. Bankrupt and delisted firms drop out of most databases, so a back-test that “buys cheap stocks” silently excludes the cheap stocks that went to zero — inflating the measured return of exactly the strategies (value, contrarian, distressed) most exposed to failures.

**Definition 7 (data-mining / data-snooping bias)** **Data mining** is finding a pattern that fits **past** data by chance because you tested enough patterns. Test 100 signals at the 5% significance level and roughly 5 will look “significant” on noise alone. The more variants you try, the more certainly your “best” back-test is fitted to history and will not repeat.

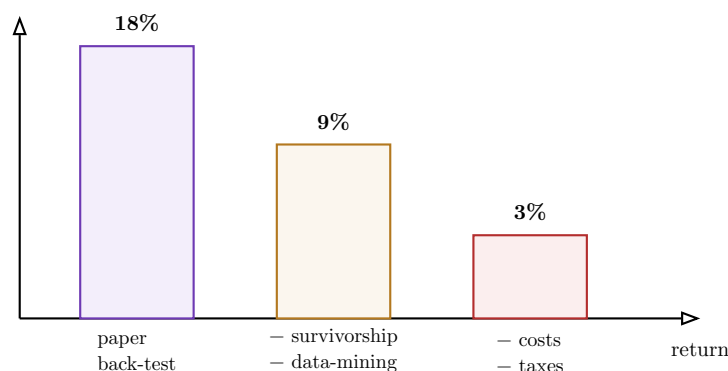


Figure 2: The waterfall from paper return to live return. A back-test’s headline number is the **top** of a descent: survivorship and data-mining bias inflate the paper figure, and trading costs and taxes (Chapter 12) erode what remains. The discipline of learning outcome 2 is to estimate this whole descent **before** trusting any back-test.

**Common pitfall** The transaction-cost assumption is where most back-tests quietly cheat. A signal that fires on **small, illiquid** stocks may show huge paper returns that vanish the moment you include the bid-ask spread and price impact of actually buying them at scale. **Always ask of a back-test: at what size, in what stocks, and with what assumed cost?** A strategy that only works in micro-caps at zero cost is not a strategy.

### Exercises

1. Take a published “anomaly” back-test and identify, specifically, how survivorship bias, data-mining risk, and the transaction-cost assumption could each be inflating its reported return. Estimate a haircut for each.
2. Classify three real strategies (a momentum rule, a low-P/E screen, an insider-following rule) by which form of EMH each implicitly bets against.

## 3 Value investing

### 3.1 Paying 60 cents for a dollar

A holding company trades at a market capitalisation of \$8bn, yet it owns a 30% stake in a listed firm worth \$11bn and has no debt. You can buy, through the holding company, more than a dollar of visible value for sixty cents — and you are paid to wait for the gap to close. This is the value investor’s archetypal situation: **price has detached from a conservatively estimated intrinsic value, and the gap is your margin of safety**. The hard part is never the arithmetic; it is telling this from a **value trap**, where the sixty cents is exactly what the dollar is now worth.

### 3.2 Graham’s principles and the margin of safety

**Definition 8 (value investing)** Value investing buys assets trading **below a conservative estimate of intrinsic value** and waits for price to converge to value. It bets the **semi-strong** EMH fails: that public information is sometimes mispriced because other investors are impatient, fearful, or indifferent to unglamorous businesses.

**Definition 9 (margin of safety)** The **margin of safety** is the gap between price and conservatively estimated intrinsic value. It is value investing’s central risk control: buying far enough below value that **even if your estimate is too optimistic**, you still do not overpay. The larger the uncertainty in the business, the larger the margin you demand.

**Definition 10 (Mr. Market)** Graham’s parable: imagine a manic-depressive partner, **Mr. Market**, who every day quotes a price to buy your share or sell you his. Some days he is euphoric and quotes absurdly high; some days despairing and quotes absurdly low. His mood is your **opportunity**, not your guide — you transact only when his price diverges from value, and ignore him otherwise.

### 3.3 Buffett’s evolution: from cigar butts to moats

**Worked example — The shift that defines modern value.** Early Buffett, following Graham, bought **statistically cheap** “cigar butts” — mediocre businesses trading below liquidation value, good for “one last puff.” Influenced by Charlie Munger, he evolved to “**a wonderful business at a fair price**” over a fair business at a wonderful price: paying up for durable competitive advantage (a **moat**), high ROIC, and pricing power, then holding for years. The lesson is that **cheapness alone is not value** — a cheap business that compounds at a low ROIC destroys value while you wait. Quality is part of the value equation, not separate from it.



### 3.4 The classic screens — and their failure mode

**Definition 11 (the three value screens)** Value candidates are screened on:

- **Low P/E** (price  $\div$  earnings) — cheap relative to current profits.
- **Low P/B** (price  $\div$  book value) — cheap relative to net assets; Graham's favourite.
- **High dividend yield** (dividend  $\div$  price) — cheap relative to cash returned.

A screen is only the **first filter**; it surfaces candidates, it does not pick winners.

**Definition 12 (value trap)** A **value trap** is a stock that is **statistically** cheap and **stays** cheap — or gets cheaper — because the low multiple correctly reflects deteriorating fundamentals: secular decline, eroding moat, value-destroying management, or accounting that flatters past earnings. The multiple is low for a **reason**, and that reason is permanent.

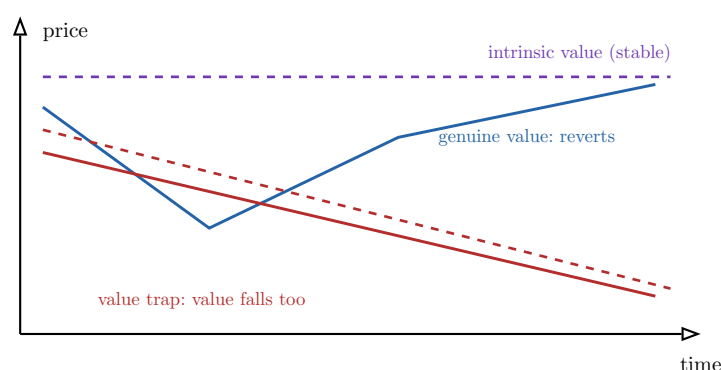


Figure 3: Genuine value versus a value trap. In genuine value (blue) intrinsic value is stable and price reverts up to it. In a value trap (red) the low multiple is correct **because intrinsic value itself is declining** — price stays cheap or falls because the business is worth less every year.

The screen cannot tell these apart; only a judgement about **why** the multiple is low can.

**Common pitfall** A screen finds cheapness; it cannot find the reason for it. Every value screen is dominated by value traps, because deteriorating businesses are mechanically the cheapest. The entire skill of value investing is the second step the screen cannot do: deciding whether the low multiple reflects **temporary mispricing** (buy) or **permanent impairment** (avoid). Filter the screen output for **quality** — durable ROIC, intact moat, honest accounting, a catalyst for re-rating — before you buy anything.

#### Exercises

3. Run low-P/E, low-P/B, and high-dividend-yield screens on a stock universe, then filter the output for quality and articulate, for each survivor, what distinguishes a value opportunity from a value trap.
4. For a named cheap stock, estimate intrinsic value conservatively, state your margin of safety, and identify the single fundamental fact that, if true, would make it a trap.

## 4 Contrarian investing

### 4.1 Buying what everyone is selling

After a sector-wide panic, the five worst-performing large caps of the year are down 50–70%. The contrarian thesis is uncomfortable and specific: **the market overreacts to bad news**, pushing prices below value, and a portfolio of the biggest **losers** will subsequently outperform the biggest **winners**. This is close to value investing but the trigger is different — value buys on a **low multiple**, contrarian buys on a **severe recent price decline**.

### 4.2 The overreaction hypothesis

**Definition 13 (contrarian investing / overreaction hypothesis)** Contrarian investing buys assets the market has **excessively punished** and (sometimes) sells those it has excessively rewarded, betting on **mean reversion**. The **overreaction hypothesis** (De Bondt–Thaler) holds that investors over-extrapolate recent bad news, so prior “losers” become underpriced and, over a multi-year horizon, **outperform** prior “winners.”

**Definition 14 (overreaction vs rational repricing)** The central distinction. A price fall is:

- **Overreaction** if the market has pushed price below the **unchanged** intrinsic value out of fear — this reverts, and is the contrarian’s opportunity.
- **Rational repricing** if intrinsic value **genuinely fell** (a structural shock, a broken business model) — this does **not** revert, and buying it is catching a falling knife.

**Common pitfall** Contrarian investing and value traps are the **same danger** wearing different clothes. A stock down 70% is either an overreaction (buy) or a business whose value really collapsed (avoid) — and the market crash that creates the opportunity is exactly when the two are hardest to tell apart. The contrarian’s discipline is to demand evidence that **value is unchanged** (intact balance sheet, durable demand, temporary cause) before equating a price fall with mispricing. “It fell a lot” is not a thesis.

### 4.3 The horizon and the cost problem

**Remark** Contrarian reversals play out over **years**, not quarters — and the position often falls **further** before it reverts. This demands a long horizon and a tolerance for being wrong for a long time before being right. It also generates real costs: you are buying illiquid, distressed names where spreads are wide, and you may rebalance into deeper losers repeatedly. Estimate these costs and confirm the horizon **before** committing; a contrarian strategy on a short horizon is just buying losers.

#### Exercises

5. Evaluate a contrarian thesis on a stock down sharply: marshal the evidence that distinguishes genuine overreaction from rational repricing, state the horizon required, and estimate realistic transaction costs.

## 5 Growth investing and GARP

### 5.1 When a high multiple is cheap

A software firm trades at a P/E of 40 — expensive by any value screen. But earnings are growing 35% a year with a long runway. Is it overvalued? A pure value screen says yes reflexively; a growth investor says **it depends on whether the growth justifies the price**. The tool that makes this judgement is the **PEG ratio**, and the philosophy that uses it disciplined is **GARP**.

### 5.2 Growth, GARP, and the PEG ratio

**Definition 15 (growth investing)** Growth investing buys companies expected to grow earnings **faster than the market**, accepting high current multiples in exchange for future earnings power. It bets that the market **underestimates the magnitude or durability of growth**.

**Definition 16 (GARP and the PEG ratio)** GARP (growth at a reasonable price) is the disciplined middle ground: buy growth, but only when its price is justified by its rate. The key metric is the **PEG ratio**:

$$\text{PEG} = \frac{\text{P/E ratio}}{\text{annual earnings growth rate (\%)}}$$

A rough rule of thumb: **PEG** < 1 suggests growth is cheaply priced; **PEG** > 1 suggests you may be overpaying. A P/E of 40 with 40% growth gives PEG = 1; the same P/E with 10% growth gives PEG = 4 — the same multiple, a very different bet.

**Worked example — Two firms, same P/E, opposite verdicts.** Firm A: P/E 30, growth 30% → PEG = 1.0. Firm B: P/E 30, growth 8% → PEG = 3.75. The value screen flags both as “expensive at 30×.” GARP says A is **reasonably priced for its growth** and B is **dangerously expensive**. The discipline is that the multiple alone is meaningless; only the multiple **relative to growth** carries information.

### 5.3 Why pure growth investing fails

**Definition 17 (growth sustainability)** The fragile assumption in any growth thesis is that the high growth rate **persists** long enough to justify the multiple. Empirically, **high growth rates mean-revert fast**: competition, market saturation, and the law of large numbers drag exceptional growers toward the average within a few years. The PEG ratio is only as good as the growth forecast in its denominator — and that forecast is usually too high and too long.

**Common pitfall** Overpaying for growth is exactly as dangerous as buying a value trap — the symmetry is the point. The value trap is a cheap stock whose earnings keep falling; the **growth trap** is an expensive stock whose growth decelerates faster than priced, collapsing the multiple **and** the earnings estimate at once. “Pure” growth investing — paying any price for growth, ignoring valuation — fails because growth is **mean-reverting and**

**over-forecast**, so the highest-multiple names are systematically the most disappointing. GARP's "reasonable price" is the discipline that pure growth lacks.

### Exercises

6. Calculate PEG ratios for a set of growth stocks, assess the **sustainability** of each growth rate (competition, saturation, base effects), and identify which are GARP buys and which are growth traps.
7. Explain, using a concrete example, why overpaying for growth and buying a value trap are symmetric errors.

## 6 Momentum and mean reversion

### 6.1 The pattern that should not exist

Sort stocks by their return over the past 12 months. Buy the top decile, short the bottom decile, hold 6–12 months. Historically this **works** — past winners keep winning and past losers keep losing over the medium term. It directly contradicts the **weak-form** EMH, which says past prices carry no information. Yet over **longer** horizons the opposite holds: 3–5-year winners tend to **reverse**. The same market exhibits **momentum** in the medium term and **reversal** in the long term.

### 6.2 Three horizons, three behaviours

**Definition 18 (momentum and mean reversion across horizons)** Returns show different autocorrelation at different horizons:

- **Short term (days–weeks)**: weak, noisy reversal (bid-ask bounce, liquidity effects) — hard to trade net of cost.
- **Medium term (3–12 months)**: **momentum** — winners keep winning, losers keep losing.
- **Long term (3–5 years)**: **mean reversion** / **reversal** — extreme past winners underperform, past losers outperform (the contrarian effect of Chapter 4).

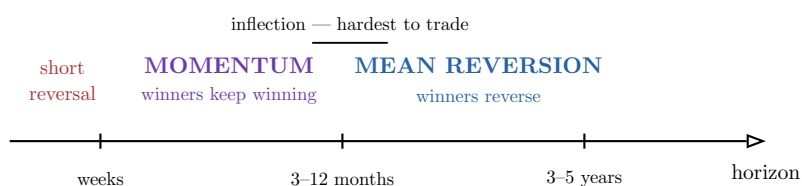


Figure 4: The same market is **trending** over months and **mean-reverting** over years. The money is hardest to make at the **inflection points** where momentum turns into reversal — precisely where the trend that made the profit ends and reverses.

### 6.3 Why the profits are so hard to capture

**Common pitfall** Momentum exists in the data but is **brutal to harvest live**. Three reasons. First, **high turnover** — the winner set changes constantly, so momentum strategies trade a lot and pay heavy transaction costs (Chapter 12) that eat much of the paper return. Second, **crashes** — momentum suffers rare, violent reversals (e.g. sharp rebounds off market

bottoms) that wipe out months of gains in days. Third, **the inflection problem** — the profit depends on exiting **before** the trend reverses, but the reversal is only obvious in hindsight; you systematically give back the turn. A momentum back-test that ignores turnover cost and crash risk is one of the most misleading objects in finance.

### Exercises

8. Describe the empirical momentum and mean-reversion patterns across the three horizons, cite the evidence for each, and explain — with reference to turnover, costs, and inflection points — why momentum profits are hard to capture in practice.

## 7 Information-based strategies

### 7.1 Trading on who knows what, and how fast

Three traders watch the same blow-out earnings report. One bought **before** it on private knowledge; one buys **at** the announcement; one buys the **week after**, betting the price has not yet fully adjusted. Whether each makes money is a precise question about **what information is in the price and how fast it gets there** — the empirical heart of market efficiency, applied to trading.

### 7.2 Insider patterns, analyst bias, and PEAD

#### Definition 19 (the three information signals)

- **Insider trading patterns** — legal insider transactions (executives buying/selling their own firm, disclosed in filings) predict returns somewhat: insider **buying** is a mild positive signal. (**Illegal** trading on material non-public information is a strong signal but a crime — the strong-form EMH fails, but you cannot legally exploit it.)
- **Analyst recommendation bias** — sell-side analysts skew **optimistic** (few “sell” ratings) because of conflicts: investment-banking relationships, access to management, career incentives. The **level** of a rating is noisy; **changes** and **deviations from consensus** carry more information.
- **Post-earnings-announcement drift (PEAD)** — after a large earnings surprise, the price keeps **drifting** in the surprise’s direction for weeks. A direct violation of the semi-strong EMH: public information is not fully impounded immediately.

**Worked example — PEAD as the cleanest anomaly.** A firm reports earnings 20% above consensus. The price jumps on the day — but then drifts **further up** over the following 60 days instead of settling. Buying the largest positive surprises and shorting the largest negative ones has earned abnormal returns for decades. PEAD is among the most robust, most replicated anomalies — and **still** the practical question is whether the drift exceeds the cost of trading it.

### 7.3 Which advantages survive net of cost

**Common pitfall** An information edge is only real if it survives **two** tests, not one. First, **is it legal and exploitable?** — strong-form inefficiency from inside information is real but criminal, so irrelevant to a legitimate strategy. Second, **does it survive cost?** — many documented signals (including parts of PEAD) live disproportionately in small, illiquid stocks

where spreads and price impact consume the abnormal return. The professional question is never “does this signal predict returns?” but “does this signal predict returns **net of the cost of trading it at my size?**” Most do not.

### Exercises

9. Evaluate three information-based strategies — following legal insider buying, fading analyst optimism, trading PEAD — and assess, for each, whether the advantage survives net of transaction cost and legal constraint.

## 8 Activist investing

### 8.1 Buying a vote, not just a share

A conglomerate trades far below the sum of its parts: a great consumer brand bolted to a cash-draining industrial division, run by an entrenched board. A passive value investor buys and **hopes** the gap closes. An **activist** buys a large stake and **forces** it to close — pushing to sell the industrial division, return cash, and replace directors. The activist’s edge is not just identifying the mispricing; it is **being the catalyst** that realises it.

### 8.2 How activists create value

**Definition 20 (activist investing)** **Activist investing** takes a significant stake in a company and uses the rights of ownership — board seats, shareholder proposals, public campaigns, proxy fights — to **force changes** that unlock value. The activist supplies the **catalyst** a passive value position lacks. The thesis: the business is mispriced **because of fixable choices** — poor capital allocation, a bloated cost base, a value-destroying conglomerate structure, weak governance.

**Definition 21 (the activist levers)** Activists typically pull one or more of:

- **Capital structure** — push for buybacks, dividends, or more leverage to return idle cash.
- **Asset mix** — spin off or sell divisions so a conglomerate’s parts are valued separately (closing the **conglomerate discount**).
- **Governance and management** — replace directors or the CEO, change incentive structures, improve disclosure.
- **Operations / strategy** — cut costs, exit unprofitable lines, redirect strategy.

### 8.3 When activism destroys value

**Common pitfall** Activism creates value when it **corrects a genuine governance or allocation failure** — but it **destroys** value when it forces **short-term** financial engineering at the expense of the business: loading a cyclical firm with debt for a one-off buyback, cutting R&D or capex the franchise needs to hit a near-term EPS target, or breaking up a group whose parts were genuinely worth more together. The distinction is **time horizon**: value-creating activism fixes a durable problem; value-destroying activism extracts cash today and leaves a weaker company tomorrow. Judge a campaign by **what it does to the earnings power of the business**, not the pop in the share price on announcement day.

## Exercises

10. Analyse a real activist campaign: explain how the activist identified the target, which levers (capital structure, asset mix, governance, operations) were pulled, and assess — on a multi-year horizon — whether the campaign created or destroyed value.

## 9 Arbitrage strategies — and the LTCM lesson

### 9.1 Three things called “arbitrage”

The word “arbitrage” is used for three very different activities, and conflating them is how sophisticated investors blow up. The same bond trading at two prices in two markets; two share classes of one company that must eventually converge; a takeover target trading below the offer price. Each is called arbitrage; only the first is riskless.

### 9.2 The arbitrage spectrum

#### Definition 22 (the three kinds of arbitrage)

- **Pure arbitrage** — the same asset (or a replicable payoff) trades at two prices **simultaneously**; you lock in a **riskless** profit. Genuinely riskless, therefore rare, tiny, and instantly competed away.
- **Near arbitrage** — assets that are **almost** identical and **should** converge (dual-listed shares, on-the-run vs off-the-run bonds), but convergence is **not guaranteed by a fixed date**. The risk is **convergence risk**: the gap can widen before it closes, and you may be forced out first.
- **Speculative arbitrage** — a bet on a **probable** convergence that is genuinely uncertain. The archetype is **merger arbitrage**: buy the target below the offer, earn the spread **if** the deal closes — and take a large loss if it breaks.

1. leverage magnifies the convergence/break risk on the right

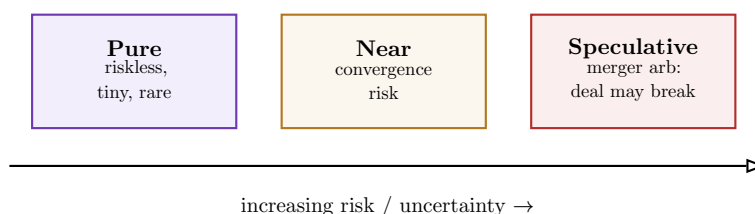


Figure 5: The arbitrage spectrum. “Riskless” applies only to the leftmost box; moving right, the convergence becomes less certain and the strategy becomes a **bet**. Leverage (next section) is what turns a small, measurable risk on this spectrum into an existential one.

### 9.3 LTCM: leverage turns measurable risk into unmeasurable uncertainty

**Definition 23 (risk versus uncertainty (Knight))** **Risk** is a hazard you can **quantify** — a known distribution of outcomes you can size and hedge. **Uncertainty** is a hazard you **cannot** quantify — the model itself may be wrong, or the world may move outside the data the model was built on. The central lesson of LTCM is that **leverage transforms the first into the second**.



**Worked example — Long-Term Capital Management, 1998.** LTCM ran near-arbitrage convergence trades — tiny, well-measured spreads between almost-identical bonds — that were genuinely low-risk on any normal distribution. To turn tiny spreads into large returns, they applied **enormous leverage** (balance-sheet leverage of 25:1, far more through derivatives). When the 1998 Russian default triggered a global flight to liquidity, spreads that “should” converge **widened** sharply and **all at once** across supposedly independent trades. Leverage meant the mark-to-market losses forced liquidation **before** the trades could converge — and the correlations the model treated as low spiked to one. A portfolio of measurable risks had become a single unmeasurable uncertainty. The fund lost \$4.6bn in months and required an orchestrated bailout.

**Common pitfall** **Leverage reclassifies a strategy.** The same convergence trade is **arbitrage** unlevered and **speculation** at 25:1, because leverage adds two risks the underlying trade did not have: (1) you can be **forced to liquidate** at the worst moment by margin calls, before convergence; and (2) diversification **fails when you need it** — in a crisis, independent trades become correlated as everyone delevers at once. Ask of any “arbitrage”: **what happens if the gap widens before it closes, and am I leveraged enough to be carried out first?** If the answer is yes, it is not arbitrage.

### Exercises

11. For three real situations, classify each as pure, near, or speculative arbitrage, and identify the specific risk that makes it not riskless.
12. Use LTCM to explain how leverage transforms measurable risk into unmeasurable uncertainty, and state the leverage level at which a given convergence trade should be reclassified as speculation.

## 10 Market timing

### 10.1 The cost of being out

An investor, convinced a crash is coming, sells to cash and waits to re-enter “when it’s safe.” Over the next twenty years the market’s **entire** return is concentrated in a handful of explosive days — and those days cluster **near the bottoms**, exactly when a timer is most likely to be out. Missing the best 10 days over two decades can cut a buy-and-hold return roughly in half; missing the best 20 can erase most of it. This is the central, brutal arithmetic of market timing.

### 10.2 Why timing is so seductive and so hard

**Definition 24 (market timing)** **Market timing** shifts capital between asset classes (typically equities and cash/bonds) to be exposed when markets rise and out when they fall. It requires being right **twice** — when to exit **and** when to re-enter — and it bets that the **aggregate** market is predictably mispriced, the hardest form of inefficiency to exploit.



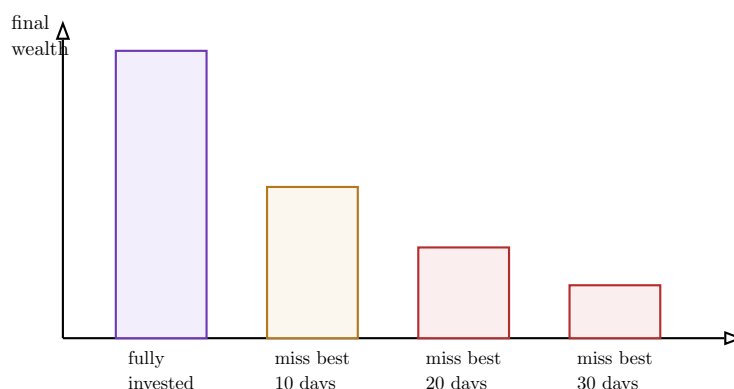


Figure 6: The documented cost of being out of the market. Because returns are concentrated in a few explosive days that **cluster near bottoms** — when a timer is most likely to have fled to cash — missing even a handful of the best days devastates long-run wealth. The timer must avoid the worst days **without** missing the best, but the two sit side by side.

## 10.3 Mean-reversion approaches and why most timing fails

**Remark** Not all timing is equal. **Valuation-based** mean-reversion timing — reducing equity exposure when broad-market valuations (e.g. a cyclically-adjusted P/E) are historically extreme — has **some** long-horizon evidence behind it, far more than chart-based or macro-forecast timing. But even it works only over very long horizons, gives weak and infrequent signals, and can be “wrong” (out of an expensive market that keeps rising) for years — long enough to break most investors’ discipline.

**Common pitfall** Most market timing fails for a compound reason: you must be right **twice** (exit and re-entry), the best and worst days **cluster together** near turning points, the strategy generates **taxes and costs** on every switch, and the **psychology** is backwards — timing feels safest (go to cash) exactly at bottoms and feels safe (stay invested) exactly at tops. The honest conclusion the evidence supports: for almost all investors, **time in the market beats timing the market**, and the only defensible timing is mild, valuation-anchored, and rare.

### Exercises

**13.** Using the cost-of-being-out evidence, evaluate a proposed market-timing strategy: quantify what it must get right, assess any mean-reversion signal it relies on, and articulate why most timing strategies fail.

## 11 The case for passive investing

### 11.1 The arithmetic the average active manager cannot beat

Before any back-test, one fact constrains the whole active-versus-passive debate. The **average** actively managed dollar must, before costs, earn the **market** return — because active investors collectively **are** the market. After their higher fees and trading costs, the average active dollar must therefore earn **less** than the market. This is not an empirical finding to be debated; it is **arithmetic** (Sharpe’s “arithmetic of active management”). The empirical question is only

**how much** worse, and the answer is: most active funds underperform their benchmark over long horizons, and the minority that win in one period rarely repeat.

## 11.2 Index and ETF mechanics

**Definition 25 (passive investing)** **Passive investing** abandons the attempt to beat the market and instead **owns the market** cheaply, through index funds or ETFs that replicate a benchmark. It is the rational default if you believe markets are efficient enough that, after costs, your active edge is negative — the conclusion most investors should reach about themselves.

### Definition 26 (index funds and ETFs)

- **Index fund** — a mutual fund holding the benchmark's constituents in benchmark weights; priced once daily at NAV. Very low turnover, very low fee.
- **ETF (exchange-traded fund)** — an index fund that **trades intraday like a stock**. Its **creation/redemption** mechanism — authorised participants exchange baskets of the underlying shares for ETF units and vice versa — keeps the ETF price tied to NAV and makes ETFs structurally **tax-efficient** (in-kind redemptions avoid realising gains).

## 11.3 When active retains a defensible edge

**Remark** Passive is the default, **not** a universal law. Active management retains a defensible edge under specific, namable conditions: **(1) genuinely inefficient markets** — illiquid, under-researched corners (micro-caps, distressed debt, frontier markets, private assets) where prices are not well discovered; **(2) concentrated, durable expertise** — a real informational or analytical advantage in a narrow domain, not a general “we pick good stocks”; **(3) specific structural advantages** — patient permanent capital that can hold through drawdowns others cannot, or a mandate to act as a catalyst (activism, Chapter 8). Absent at least one of these, the arithmetic wins and passive is correct.

**Common pitfall** The honest test before going active: **name your edge, and name why it persists**. “I’m a good analyst” is not an edge — everyone believes that, and the arithmetic already accounts for it. A defensible edge is specific (a market segment, an information source, a structural advantage) **and** durable (a reason it is not already arbitrated away). If you cannot state both, the evidence says own the index.

### Exercises

**14.** Present the empirical case for passive investing — active-fund underperformance and index/ETF mechanics — then name the specific conditions under which active management retains a defensible edge, and test a named active fund against them.

## 12 Trading costs and taxes: the hidden drag

### 12.1 The return you keep, not the return you earn

Two strategies both earn 10% a year **gross**. The first trades twice a year in liquid large caps; the second trades 200% of the portfolio annually in small caps and is run in a taxable account. After spreads, price impact, and short-term capital-gains tax, the first investor keeps 9% and the second keeps 4% — and over thirty years that gap turns into a **threefold** difference in final wealth. Costs and taxes are not a footnote to a strategy; for high-turnover strategies they are often the **whole** story of why the back-test never showed up in the account.

### 12.2 The components of trading cost

#### Definition 27 (the four components of trading cost)

- **Bid-ask spread** — the gap between the price you can buy at and sell at; paid on every round trip, wider in illiquid stocks.
- **Price impact** — your own buying **pushes the price up** (and selling pushes it down); the larger your order relative to volume, the worse. This is why a strategy that works at \$10m may fail at \$10bn.
- **Commissions / fees** — explicit per-trade charges and platform fees; now small for equities but not zero.
- **Opportunity cost** — the return missed while waiting to execute, or on trades not done because the market moved away.

**Common pitfall** Cost scales with turnover and shrinks with liquidity — and these interact viciously with strategy choice. The very strategies with the strongest paper anomalies (momentum, contrarian, small-cap value, PEAD) are **high-turnover** and often live in **illiquid** stocks — exactly the combination that maximises both spread and price impact. This is the deepest reason paper returns do not survive: the signal and the cost are **correlated**, so the strongest-looking anomalies are frequently the most expensive to trade and the first to vanish net of cost.

### 12.3 Taxes and the power of deferral

**Definition 28 (tax efficiency)** Tax efficiency is structuring a strategy to minimise and defer tax. Two levers dominate: (1) **short-term vs long-term gains** — short-term gains (on positions held briefly) are typically taxed far more heavily than long-term, so **high turnover is doubly costly** — it pays more tax **and** the higher rate; (2) **deferral** — an unrealised gain compounds on the **whole** pre-tax amount, so a low-turnover buy-and-hold strategy lets the government's share keep working for you until you sell. Tax deferred is tax reduced, because of compounding.

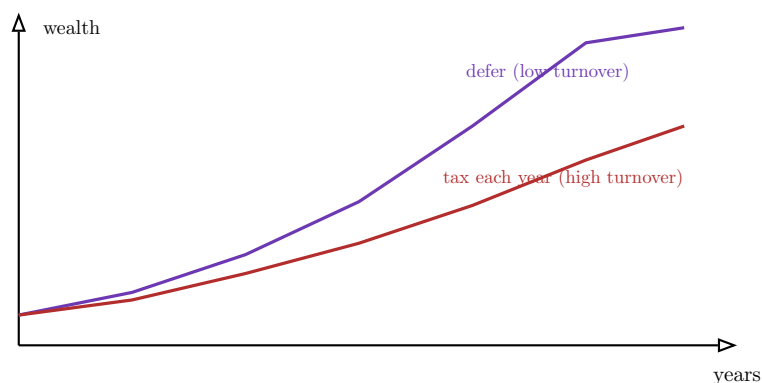


Figure 7: Why deferral compounds. Realising and taxing gains every year (red) shrinks the base that compounds; deferring tax until a single sale at the end (green) lets the whole pre-tax amount compound for decades. The gap widens **non-linearly** over time — the hidden drag of high turnover is largest precisely for long-horizon investors.

### Exercises

15. Calculate the all-in cost of a strategy — bid-ask spread, price impact, commissions, management fees, and the tax drag from its turnover — and express it as an annual haircut to gross return.
16. Show, with a multi-decade compounding example, how a 2-3% annual cost-and-tax drag erodes final wealth, and explain why high-turnover strategies are penalised twice.

## 13 Constructing and defending a personal investment philosophy

### 13.1 From eleven philosophies to one worldview

You have now tested every major philosophy against the same three filters — **evidence, cost, bias**. None is universally right; each works only under specific conditions, for specific investors, against specific inefficiencies. The course's final task is therefore not to pick a winner from the menu but to **construct your own coherent worldview** and defend it before a practitioner jury — the assessment that anchors learning outcome 1.

### 13.2 The structure of a defensible philosophy

**Definition 29 (a complete investment philosophy)** A defensible personal philosophy states, explicitly and consistently:

1. **Belief** — where and why you think markets are inefficient (which EMH form fails, and the behavioural or structural reason it persists).
2. **Edge** — why **you** can exploit it: your specific informational, analytical, structural, or temperamental advantage — and **why it is durable** (not already arbitrated away).
3. **Fit** — coherence with your **risk tolerance, time horizon, and tax status**: a strategy you can actually hold through, on a horizon you actually have, that survives your tax situation.
4. **Strategy** — the concrete, repeatable rules that implement the belief.
5. **Cost realism** — an honest estimate of transaction costs and taxes, proving the edge survives net of them.

6. **Falsification** — what evidence would tell you the philosophy is **wrong** (not just a bad year) — the discipline that lets you hold through drawdowns without holding through ruin.

**Worked example — A coherent philosophy, stated.** “I believe the semi-strong EMH fails for **unglamorous mid-caps** that fall out of analyst coverage, because institutional mandates and career incentives push attention to large, exciting names (**belief**). My edge is patience and a narrow circle of competence in industrial businesses, plus permanent capital that need not sell in drawdowns (**edge/fit**). I screen on low EV/EBIT, filter hard for durable ROIC and honest accounting to avoid value traps, and hold 3-5 years (**strategy**). Turnover is low, so costs and taxes are minimal and deferred (**cost realism**). I am wrong if these names systematically **stay** cheap because their moats are eroding — if my ‘overreactions’ turn out to be rational repricings across the portfolio (**falsification**).” Every piece connects; remove one and the worldview is incoherent.

**Common pitfall** The jury — like any real allocator — will attack the **weakest link**, and it is almost always one of three. **No durable edge**: a belief about inefficiency with no reason **you** can harvest it that the market has not already competed away. **No coherence with the investor**: a strategy whose horizon, drawdowns, or turnover the stated investor cannot actually bear. **No cost realism**: a paper edge that dies net of spread, impact, and tax. Before you defend, attack your own philosophy on these three points — because the jury will, and a worldview that cannot survive the attack was never a philosophy, just a preference.

### Exercises

17. Construct your personal investment philosophy stating all six elements — belief, edge, fit, strategy, cost realism, falsification — coherent with a specified risk tolerance, time horizon, and tax status.
18. Defend it against the three standard attacks (no durable edge, no coherence with the investor, no cost realism), anchoring every answer in course evidence.
19. Take an investor profile **different** from your own (e.g. a 65-year-old taxable retiree, or a tax-exempt 25-year endowment) and argue which philosophy best fits it and why yours would **not**.

**Investment philosophies, in one paragraph.** A philosophy is a **coherent belief about how markets work**, from which a strategy follows; you need one because its absence makes you switch approaches at the worst moments. Every active philosophy is a bet that some form of the **EMH** fails — weak (momentum), semi-strong (value, growth, PEAD), or strong (insider) — and every such bet must survive three filters: **evidence** (does the anomaly exist net of survivorship and data-mining bias?), **cost** (does it survive spread, impact, and tax, which correlate viciously with the strongest anomalies?), and **bias** (can you actually hold it through the drawdown?). **Value** buys below conservative intrinsic value with a margin of safety, and lives or dies on telling opportunity from **value trap**; **contrarian** buys overreaction but must distinguish it from **rational repricing**; **growth/GARP** pays up for growth only when **PEG** justifies it, and fails when growth is over-forecast; **momentum** is real but

brutal to harvest through turnover, crashes, and inflection points; **information** strategies (insider, analyst bias, PEAD) must survive legality **and** cost; **activism** supplies the catalyst value lacks but destroys value when it engineers the short term; **arbitrage** is riskless only when pure, and **LTCM** shows how **leverage turns measurable risk into unmeasurable uncertainty**; **market timing** mostly fails because the best days cluster with the worst and you must be right twice; and **passive** is the arithmetic default unless you can name a **durable edge**. The capstone is a personal philosophy — belief, edge, fit, strategy, cost realism, falsification — coherent with your risk, horizon, and taxes, and defended against the three attacks an allocator will always make. Three questions, always: **where do markets misprice, why can I durably exploit it, and does the edge survive cost?**